

Step-by-Step Mathematical Calculation: Hook Score, Trending Score, Compare Score

Generated: 2026-02-24 10:24

A) HOOK SCORE - FULL CALCULATION

Formula blocks from server/utills/hookScore.js

Given (example product):

views_recent=1200, views_previous=800, compares_recent=240, compares_previous=160, wishlist_recent=300, age_days=90

smoothing s=5, freshness constant tau=365

cohort maxima: max_views_recent=1500, max_compares_recent=300, max_wishlist_recent=500,

max_views_velocity_raw=1.2, max_compares_velocity_raw=1.0

Step A1: velocity raw values

$\text{views_velocity_raw} = \max(0, ((\text{views_recent} + s)/(\text{views_previous} + s)) - 1)$

$= \max(0, ((1200+5)/(800+5)) - 1)$

$= \max(0, (1205/805) - 1)$

$= \max(0, 1.496894410 - 1)$

$= 0.496894410$

$\text{compares_velocity_raw} = \max(0, ((\text{compares_recent} + s)/(\text{compares_previous} + s)) - 1)$

$= \max(0, ((240+5)/(160+5)) - 1)$

$= \max(0, 1.484848485 - 1)$

$= 0.484848485$

Step A2: normalize to 0..100

$\text{norm_views} = 100 * 1200 / 1500 = 80.000000$

$\text{norm_compares} = 100 * 240 / 300 = 80.000000$

$\text{norm_wishlist} = 100 * 300 / 500 = 60.000000$

$\text{norm_views_velocity} = 100 * 0.496894410 / 1.2 = 41.407867$

$\text{norm_compares_velocity} = 100 * 0.484848485 / 1.0 = 48.484848$

Step A3: buyer intent

$\text{buyer_intent} = \text{clamp}(0.55 * \text{norm_views} + 0.30 * \text{norm_compares} + 0.15 * \text{norm_wishlist}, 0, 100)$

$= \text{clamp}(0.55 * 80.000000 + 0.30 * 80.000000 + 0.15 * 60.000000, 0, 100)$

$= \text{clamp}(44.000000 + 24.000000 + 9.000000, 0, 100)$

$= 77.000000$

Step A4: trend velocity

$\text{trend_velocity} = \text{clamp}(0.70 * \text{norm_views_velocity} + 0.30 * \text{norm_compares_velocity}, 0, 100)$

$= \text{clamp}(0.70 * 41.407867 + 0.30 * 48.484848, 0, 100)$

$= \text{clamp}(28.985507 + 14.545455, 0, 100)$

$= 43.530962$

Step A5: freshness

$\text{freshness} = \text{clamp}(100 * \exp(-\text{age_days} / \text{tau}), 0, 100)$

= clamp(100 * exp(-90/365), 0, 100)
= clamp(100 * exp(-0.246575), 0, 100)
= 78.147248

Step A6: final hook score

hook_score = clamp(0.50*buyer_intent + 0.30*trend_velocity + 0.20*freshness, 0, 100)
= clamp(0.50*77.000000 + 0.30*43.530962 + 0.20*78.147248, 0, 100)
= clamp(38.500000 + 13.059289 + 15.629450, 0, 100)
= 67.188738

B) TRENDING SCORE - FULL CALCULATION

Formula blocks from server/utls/trendingScore.js

Given for same product (7d windows):

views_7d=1200, views_prev_7d=800, compares_7d=240, smoothing s=5
product_type partition maxima: max_views_7d=2000, max_compares_7d=400, max_velocity_raw=1.0

Step B1: velocity_raw

velocity_raw = max(0, ((views_7d + s)/(views_prev_7d + s)) - 1)
= max(0, ((1200+5)/(800+5)) - 1)
= 0.496894410

Step B2: normalize

norm_views = 100*1200/2000 = 60.000000
norm_compares = 100*240/400 = 60.000000
norm_velocity = 100*0.496894410/1.0 = 49.689441

Step B3: final trending score

trending_score = 0.40*norm_views + 0.40*norm_compares + 0.20*norm_velocity
= 0.40*60.000000 + 0.40*60.000000 + 0.20*49.689441
= 24.000000 + 24.000000 + 9.937888
= 57.937888

C) COMPARE SCORE - FULL CALCULATION (USING YOUR SMARTPHONES)

Formula blocks from server/utls/compareScoring.js

Default final weights: perf=0.36, display=0.20, camera=0.20, battery=0.14, priceValue=0.10

Device: realme P4 5G

Given: perf=62.0, refresh=144Hz, panel=AMOLED, main_mp=50, sensors=2, battery=7000mAh, price=19999

Step C1: display

refresh_score = round(20 + ((clamp(refresh,60,165)-60)/(165-60))*40)
= round(20 + ((144-60)/105)*40) = 52
panel_score(AMOLED) = 34
display_score = clamp(52 + 34, 0, 100) = 86.000000

Step C2: camera

megapixel_score = clamp((main_mp/108)*65,18,65)

$= \text{clamp}((50/108)*65, 18, 65) = 30.092593$
 $\text{sensor_score} = \text{clamp}(\text{sensor_count} * 8.75, 10, 35)$
 $= \text{clamp}(2 * 8.75, 10, 35) = 17.500000$
 $\text{camera_score} = \text{clamp}(30.092593 + 17.500000, 0, 100) = 47.592593$
 Step C3: battery
 $\text{battery_score}(7000) = 100$
 Step C4: base_spec
 $\text{base_spec} = 0.40 * \text{perf} + 0.20 * \text{display} + 0.25 * \text{camera} + 0.15 * \text{battery}$
 $= 0.40 * 62.000000 + 0.20 * 86.000000 + 0.25 * 47.592593 + 0.15 * 100.000000$
 $= 24.800000 + 17.200000 + 11.898148 + 15.000000$
 $= 68.898148$
 Step C5: value_raw
 $\text{value_raw} = \text{base_spec} / \text{price}$
 $= 68.898148 / 19999 = 0.003445079661$

Device: moto g57 Power 5G

Given: perf=62.0, refresh=120Hz, panel=LCD, main_mp=50, sensors=3, battery=7000mAh, price=14999

Step C1: display
 $\text{refresh_score} = \text{round}(20 + ((\text{clamp}(\text{refresh}, 60, 165) - 60) / (165 - 60)) * 40)$
 $= \text{round}(20 + ((120 - 60) / 105) * 40) = 43$
 $\text{panel_score}(\text{LCD}) = 24$
 $\text{display_score} = \text{clamp}(43 + 24, 0, 100) = 67.000000$
 Step C2: camera
 $\text{megapixel_score} = \text{clamp}((\text{main_mp} / 108) * 65, 18, 65)$
 $= \text{clamp}((50 / 108) * 65, 18, 65) = 30.092593$
 $\text{sensor_score} = \text{clamp}(\text{sensor_count} * 8.75, 10, 35)$
 $= \text{clamp}(3 * 8.75, 10, 35) = 26.250000$
 $\text{camera_score} = \text{clamp}(30.092593 + 26.250000, 0, 100) = 56.342593$
 Step C3: battery
 $\text{battery_score}(7000) = 100$
 Step C4: base_spec
 $\text{base_spec} = 0.40 * \text{perf} + 0.20 * \text{display} + 0.25 * \text{camera} + 0.15 * \text{battery}$
 $= 0.40 * 62.000000 + 0.20 * 67.000000 + 0.25 * 56.342593 + 0.15 * 100.000000$
 $= 24.800000 + 13.400000 + 14.085648 + 15.000000$
 $= 67.285648$
 Step C5: value_raw
 $\text{value_raw} = \text{base_spec} / \text{price}$
 $= 67.285648 / 14999 = 0.004486008944$

Device: Redmi 15 5G

Given: perf=62.0, refresh=144Hz, panel=DotDisplay, main_mp=50, sensors=2, battery=7000mAh, price=15999

Step C1: display
 $\text{refresh_score} = \text{round}(20 + ((\text{clamp}(\text{refresh}, 60, 165) - 60) / (165 - 60)) * 40)$
 $= \text{round}(20 + ((144 - 60) / 105) * 40) = 52$
 $\text{panel_score}(\text{DotDisplay}) = 22$
 $\text{display_score} = \text{clamp}(52 + 22, 0, 100) = 74.000000$

Step C2: camera

$\text{megapixel_score} = \text{clamp}((\text{main_mp}/108)*65, 18, 65)$
 $= \text{clamp}((50/108)*65, 18, 65) = 30.092593$
 $\text{sensor_score} = \text{clamp}(\text{sensor_count}*8.75, 10, 35)$
 $= \text{clamp}(2*8.75, 10, 35) = 17.500000$
 $\text{camera_score} = \text{clamp}(30.092593 + 17.500000, 0, 100) = 47.592593$

Step C3: battery

$\text{battery_score}(7000) = 100$

Step C4: base_spec

$\text{base_spec} = 0.40*\text{perf} + 0.20*\text{display} + 0.25*\text{camera} + 0.15*\text{battery}$
 $= 0.40*62.000000 + 0.20*74.000000 + 0.25*47.592593 + 0.15*100.000000$
 $= 24.800000 + 14.800000 + 11.898148 + 15.000000$
 $= 66.498148$

Step C5: value_raw

$\text{value_raw} = \text{base_spec} / \text{price}$
 $= 66.498148/15999 = 0.004156394034$

Step C6: normalize value to value_score

$\text{min_value_raw} = 0.003445079661$
 $\text{max_value_raw} = 0.004486008944$
 $\text{value_score} = 35 + ((\text{value_raw} - \text{min})/(\text{max}-\text{min}))*65$

realme P4 5G: $\text{value_score} = 35 + ((0.003445079661 - 0.003445079661)/(0.004486008944 - 0.003445079661))*65 = 35.000000$
moto g57 Power 5G: $\text{value_score} = 35 + ((0.004486008944 - 0.003445079661)/(0.004486008944 - 0.003445079661))*65 = 100.000000$
Redmi 15 5G: $\text{value_score} = 35 + ((0.004156394034 - 0.003445079661)/(0.004486008944 - 0.003445079661))*65 = 79.417459$

Step C7: final overall score

$\text{overall} = 0.36*\text{perf} + 0.20*\text{display} + 0.20*\text{camera} + 0.14*\text{battery} + 0.10*\text{value_score}$

realme P4 5G: $\text{overall} = 0.36*62.000000 + 0.20*86.000000 + 0.20*47.592593 + 0.14*100.000000 + 0.10*35.000000 = 66.538519$
moto g57 Power 5G: $\text{overall} = 0.36*62.000000 + 0.20*67.000000 + 0.20*56.342593 + 0.14*100.000000 + 0.10*100.000000 = 70.988519$
Redmi 15 5G: $\text{overall} = 0.36*62.000000 + 0.20*74.000000 + 0.20*47.592593 + 0.14*100.000000 + 0.10*79.417459 = 68.580264$

Step C8: ranking rule

Sort by overall DESC. Tie-breaker 1: lower price first. Tie-breaker 2: alphabetical name.

Ranking result:

1. moto g57 Power 5G (overall=70.988519, price=14999)
2. Redmi 15 5G (overall=68.580264, price=15999)
3. realme P4 5G (overall=66.538519, price=19999)

D) CONNECTION BETWEEN THE 3 ALGORITHMS

- 1) Hook and Trending are connected: both are behavior/engagement based and share views + comparisons + velocity logic.
- 2) Compare is different: it is spec+price based and computed on selected products only.
- 3) In product pages, they can coexist: Hook/Trending for discovery rank, Compare for decision rank.

E) API JSON BODIES (EXACT SHAPES)

POST /api/admin/hook-score/recompute body: {}

POST /api/admin/trending/recompute body: {}

POST /api/admin/trending/boost body:

{"product_id":124,"manual_boost":true,"manual_priority":5,"manual_badge":"editor_pick"}

PUT /api/admin/compare-scoring body:

{"weights":{"performance":36,"display":20,"camera":20,"battery":14,"priceValue":10},"chipset_rules":[{"keyword":"snapdragon 8 gen 3","score":95}]}

POST /api/public/compare/scores body:

{"devices":[{"product_id":166,"variant_id":1613},{"product_id":165,"variant_id":1605},{"product_id":159,"variant_id":1610}]}